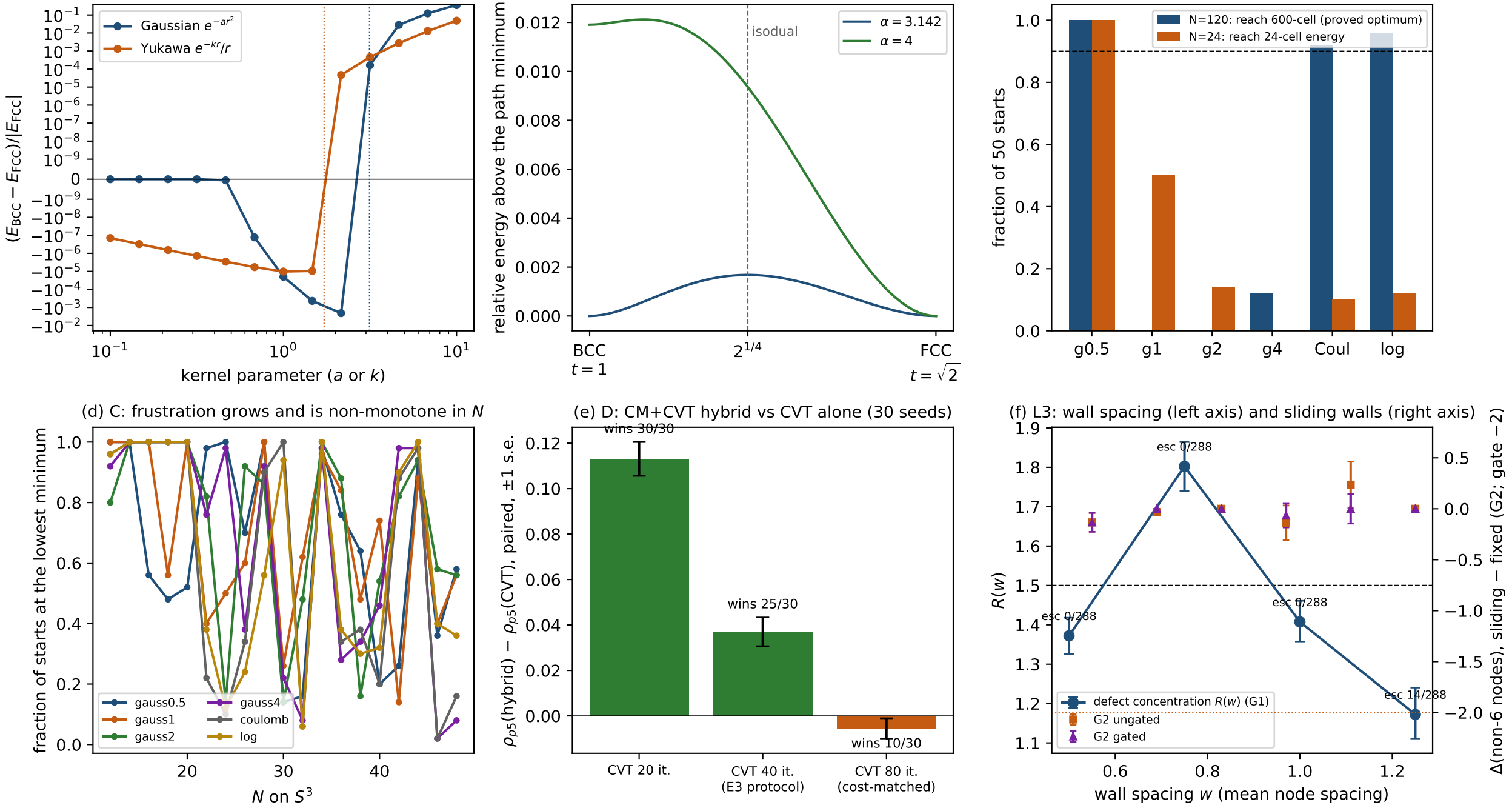
(a) A: Bravais winner; crossovers $a^* = \pi$, $k^* = 1.718$
Gaussian e^{-ar^2}
Yukawa e^{-kr}/r
$(E_{BCC} - E_{FCC})/|E_{FCC}|$
kernel parameter (a or k)
(b) Bain path: the self-dual lattice is the maximum at $\alpha = \pi$
$\alpha = 3.142$
$\alpha = 4$
isodual
relative energy above the path minimum
BCC
$t = 1$
$2^{1/4}$
FCC
$t = \sqrt{2}$
(c) C: local optimiser on S^3 (dashed: gate 0.9)
N=120: reach 600-cell (proved optimum)
N=24: reach 24-cell energy
fraction of 50 starts
g0.5
g1
g2
g4
Coul
log
(d) C: frustration grows and is non-monotone in N
fraction of starts at the lowest minimum
gauss0.5
gauss1
gauss2
gauss4
coulomb
log
N on S^3
(e) D: CM+CVT hybrid vs CVT alone (30 seeds)
wins 30/30
wins 25/30
wins 10/30
ρ_{p5}(hybrid) − ρ_{p5}(CVT), paired, ±1 s.e.
CVT 20 it.
CVT 40 it.
(E3 protocol)
CVT 80 it.
(cost-matched)
(f) L3: wall spacing (left axis) and sliding walls (right axis)
esc 0/288
esc 0/288
esc 0/288
esc 14/288
defect concentration $R(w)$ (G1)
G2 ungated
G2 gated
$R(w)$
wall spacing w (mean node spacing)
Δ(non-6 nodes), sliding − fixed (G2; gate −2)